Part: **4.5 – RESIDUAL STRENGTH**

Chapter: **4510**

Title: **RESIDUAL STRENGTH CAPABILITY**

Revision: **0.1**

Date: **10/12/2021**

1 INTRODUCTION

The residual strength is defined as the static load that a cracked component can withstand without failure. The residual strength decreases as the damage in the component increases (see Figure 1), and depends mainly on the material fracture toughness, the geometry and the damage size.

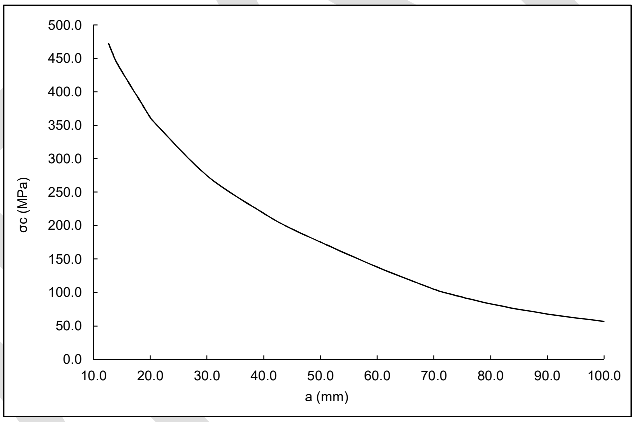


Figure 1. Evolution of the residual strength (σ_c) vs crack length (a)

The main goal of the residual strength assessment is to assure that the component has the required strength to withstand the limit load during all its service life, even when cracks are present.

The residual strength assessment involves the following steps:

- Damage tolerance analysis to obtain the relationship between the applied stress, the crack length and the stress intensity factor (covered by chapter 4420).
- Calculation of the strength that leads the structural failure for a given crack size, or calculation of the maximum allowable crack length at limit load.

This chapter is focused on the methodology to calculate the residual strength for a given crack length for metallic materials. It is worthy to mention that the behaviour around the cracked area is assumed to be governed by the basic linear fracture mechanics equations.

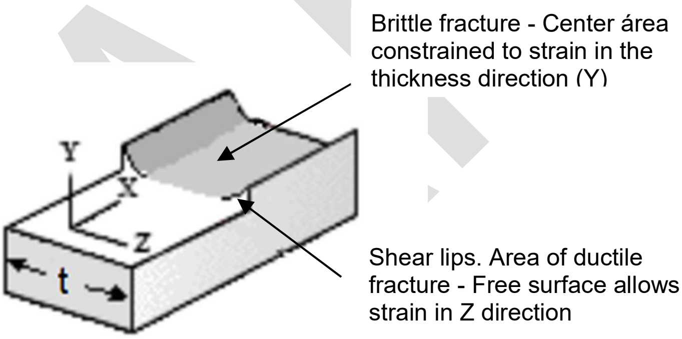
2 RESIDUAL STRENGTH ANALYSIS

In a cracked structure, the stress intensity factor (K) is the magnitude that defines the severity of the stress distribution around the crack tip. When the applied load increases, the stress intensity factor also increases.

The material fracture behaviour can be described as being ductile or brittle; brittle materials tend to a rapid and unstable crack growth while ductile materials show a slow and stable crack growth. The ductile fracture exhibits a continuous process of ductile tearing, rather than a point of fracture which is characteristic of the brittle fracture.

Different fracture behaviours require different methodologies to calculate the failure stress, depending on whether the material acts in a ductile or in a brittle manner.

Ductile fracture is predominant in thin parts (plane stress condition), due to the biaxial stress state, while thick parts (plane strain condition) tends to brittle fracture because of the triaxial stress state.



There are several criteria available. The ones here exposed are the ones recommended to be used in AD&S in the analytical calculations. The criteria to choose one or another depend on the expected crack growth behaviour and on the available material data.

Figure 2 summarizes the residual strength methodology to be followed.

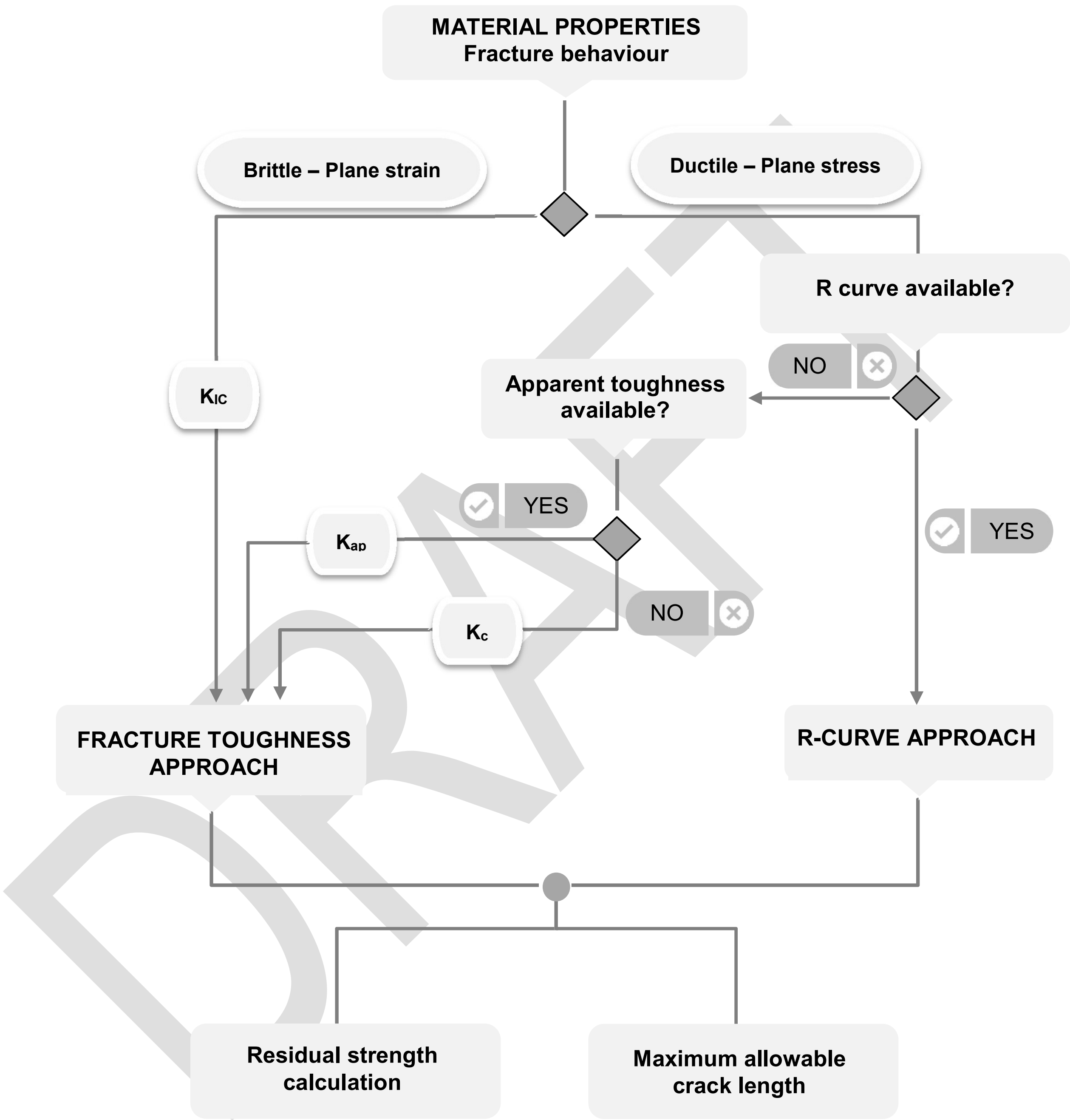


Figure 2. Residual strength calculation methodology

2.1 FRACTURE TOUGHNESS APPROACH

Abrupt fracture usually occurs in thick sheets (plane strain) or/and brittle materials. In these cases, the start of slow crack extension will be followed by the onset of rapid fracture. This approach has to be used also for part through cracks, like corner or surface cracks.

For materials that tend to abrupt fracture, the failure occurs when the stress intensity factor (K) reaches or exceeds the fracture toughness of the material ($K \geq K_{cr}$). The corresponding applied stress at failure is the residual strength (σ_c) and is calculated as:

$$\sigma_c = \frac{K_{cr}}{\beta(a_c) \cdot \sqrt{\pi \cdot a_c}}$$

The critical stress intensity factor K_{cr} is the **material fracture toughness** and it varies with the thickness. The fracture toughness is denoted as K_{IC} for plane strain condition and K_c for plane stress condition. In general, the fracture toughness decreases as thickness increases, until a plane strain condition is reached and the value of the fracture toughness stabilises (K_{IC}; plane strain fracture toughness).

The following equation is usually used to calculate the plane-strain thickness (t_0) which is considered as the transition thickness between plane-strain and plane-stress (see Ref. 2):

$$t_0 = 2.5 \cdot \left(\frac{K_{IC}}{\sigma_y}\right)^2$$

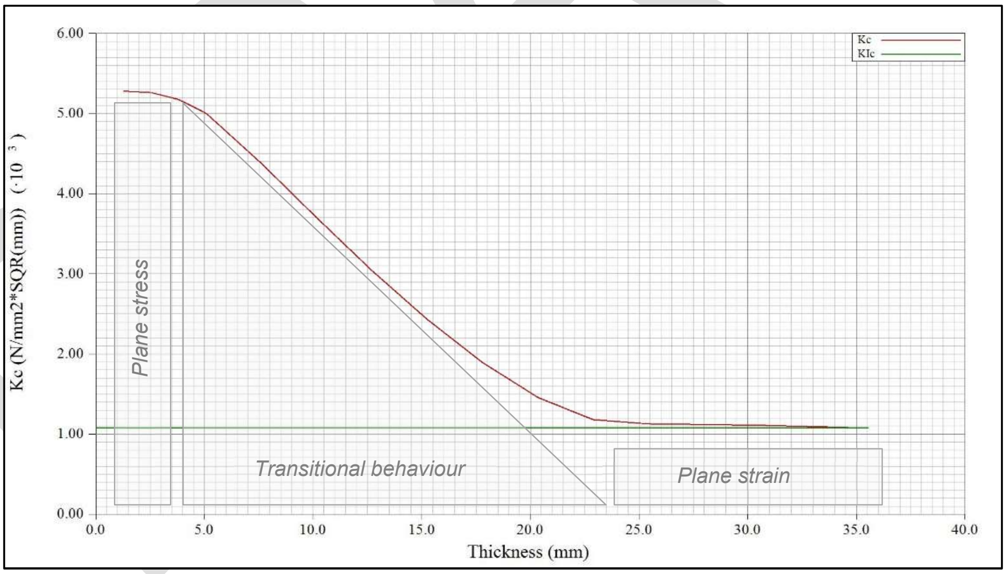


Figure 3. Fracture toughness of Aluminium 2024-T3

2.2 CRACK GROWTH RESISTANCE APPROACH

In thin metal sheets (plane stress) or/and in tough materials, the crack usually extends by a tearing mode of fracture, which leads to a stable crack growth. In this case, the failure condition depends directly on the crack growth resistance behaviour (K_R) and on the stress intensity factor for each crack length.

One of the following two analytical approaches can be used to calculate the residual strength in this case.

2.2.1 Apparent Fracture Toughness Approach

The apparent fracture toughness approach can be used by the engineer to conservatively estimate the residual strength for fast fracture of a tough material.

The apparent fracture toughness (K_{app}) is a material property that varies with the thickness (t). The curve K_{app} is obtained from the same test data employed to derive K_c, but while K_c is calculated considering the maximum stress level and the critical crack length, K_{app} is calculated as a function of the maximum stress and the initial flaw size. The residual strength (σ_c) would be calculated following the same equation as the one used in section 0:

$$\sigma_c = \frac{K_{app}}{\beta(a_c) \cdot \sqrt{\pi \cdot a_c}}$$

2.2.2 R-curve Approach

The K_R-curve measures crack resistance to tearing fracture. The R-curve describes the extent of crack movement from an initial starting condition as a function of the level of applied SIF as such represents a complete history of quasi-static crack growth up until fracture occurs (from Ref. 3).

The R-curve is usually presented as a plot of the SIF associated with a given increment of crack size, i.e. K_R, versus the effective crack increment ($\Delta a_{eff} = \Delta a + r_y$), where r_y represents the plane stress plastic zone radius. It is important to note that R-curve is usually referred to half-crack length.

Figure 4 shows the crack growth resistance curve for aluminium 2024-T3.

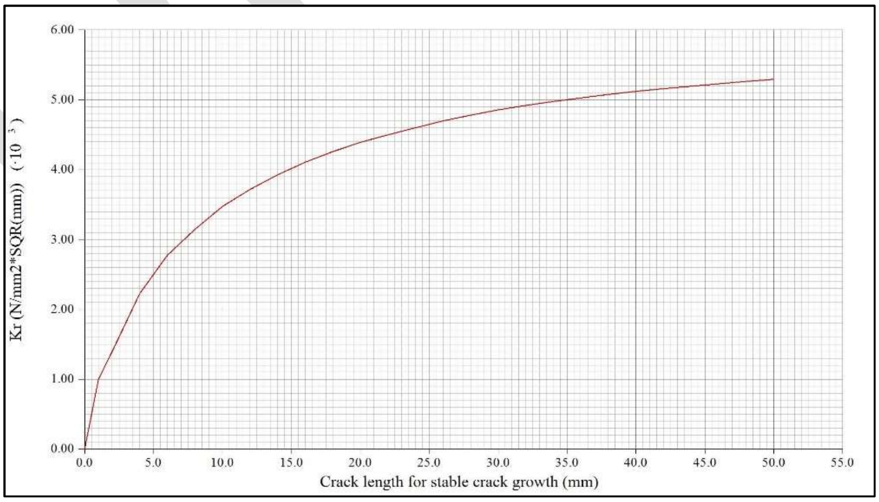


Figure 4. K_R curve for 2024-T351 (from ESDU 85031 BS L97/4)

The process to be followed to obtain the residual strength for a crack length a_{crit} is based on an iterative process in order to identify the point of tangency between the R-curve and the driving force curve, *K(σ, a)*, for the corresponding stress value:

- Position the R-curve intersecting the x-axis at a = a_{crit}.
- Calculate the stress intensity factor (SIF) for a stress σ_1:

$$K = \beta(a) \cdot \sigma_1 \cdot \sqrt{\pi \cdot a}$$

- Superimpose the driving force curve, K(σ_1, a), onto the R-curve and check if:
  - The driving force curve is below the R-curve.
  - There is a point of tangency between the driving force curve and the R-curve.
- Repeat the process at a new stress level, σ_2, until the driving force curve is below the R-curve and there is a point of tangency between both curves.

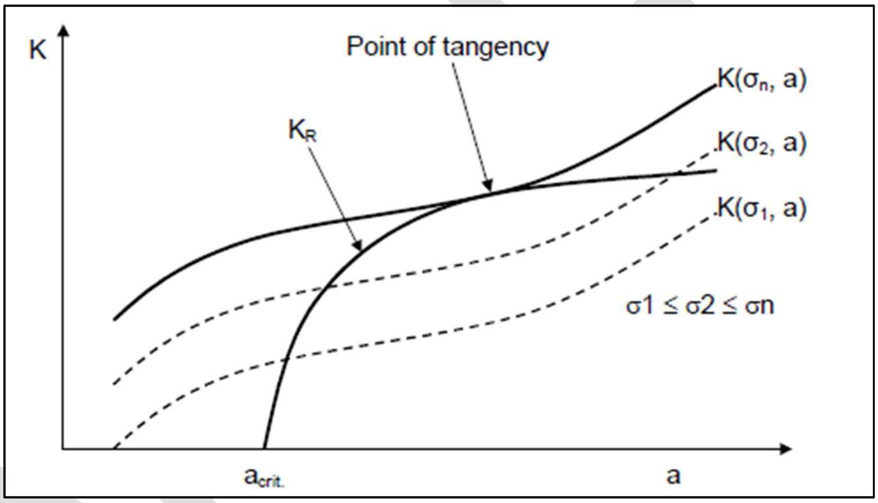


*Figure 5. Finding the residual strength using the R-curve approach (from Ref. **1**)*

Since the R-curve is calculated for the half-crack length, (*a*), the driving force curve to be used for the residual strength calculation has to be obtained taking into account the following criteria:

- For eccentric cracks and cracks emanating from holes, the maximum SIF of the two crack tips has to be considered.
- The crack length has to be calculated as:

$a_c = \frac{TipA+TipC+\emptyset}{2}$ for cracks emanating from holes

$a_c = \frac{TipA+TipC}{2}$ for eccentric cracks

$a_c = TipA$ for edge cracks

$a_c = Tip\ C$

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
a	mm	Crack length
β	-	Normalized Stress Intensity Factor
K, SIF	MPa√mm-	Stress Intensity Factor
K_C	MPa√mm	Plane stress fracture toughness
K_{IC}	MPa√mm	Plane strain fracture toughness
KR	MPa√mm	Crack growth resistance curve
σ	MPa	Stress
σ_y	MPa	Yield stress
r_y	mm	Plane stress plastic zone radius
t	mm	Thickness

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	AM2377 Module 3.1	A	Residual Strength Analysis
Ref. 2	ASTM E399	1997	Standard Test Method for Plane-Strain Fracture Toughness of Metallic Materials
Ref. 3	AFGROW	---	DTD Handbook

SOFTWARE

Program	***Version***	***Description***

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	***Authors***	***Change Reason***	***Pages/Sub-chapters affected***
0.1 10/12/2021	M. Carmen Blanquer Darío Fernández	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	***Getafe***	***Manching***
Stress Analysis (TEPAA)	Javier Gómez-Escalonilla Xx/xx/2021	Holger Hickethier Xx/xx/2021
Stress Methods (TEPAP-TL4)	Darío R. Fernández Villalba Xx/xx/2021	-

Table 3. *Approval signatures table for last revision of this chapter*